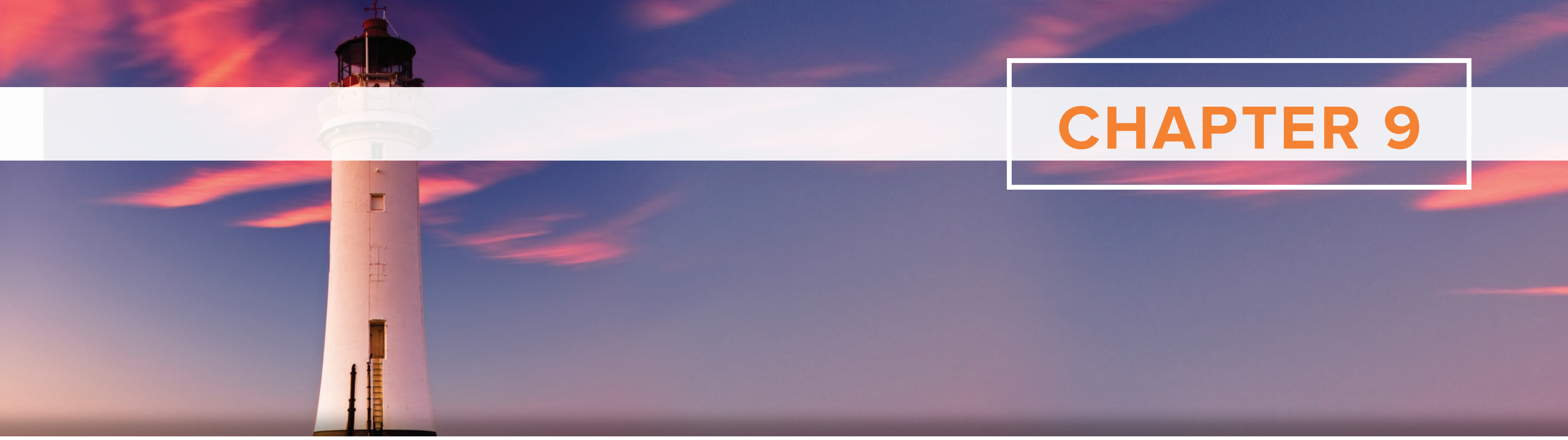

CHAPTER 9

Flexible Budgets and Performance Analysis

BUSINESS FOCUS

Why Do Companies Need Flexible Budgets?

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The difficulty of accurately predicting future financial performance can be readily understood by reading the annual report of any publicly traded company. For example, **Nucor Corporation**, a steel manufacturer headquartered in Charlotte, North Carolina, cites numerous reasons why its actual results may differ from expectations, including the following: (1) changes in the supply and cost of raw materials; (2) changes in the availability and cost of electricity and natural gas; (3) changes in the market demand for steel products; (4) fluctuations in currency conversion rates; (5) significant changes in laws or government regulations; and (6) the cyclical nature of the steel industry. ■

Source: Nucor Corporation 2014 Annual Report.

LEARNING OBJECTIVES

After studying Chapter 9, you should be able to:

LO9–1 Prepare a planning budget and a flexible budget and understand how they differ from one another.

LO9–2 Calculate and interpret activity variances.

LO9–3 Calculate and interpret revenue and spending variances.

LO9–4 Prepare a performance report that combines activity variances and revenue and spending variances.

LO9–5 Prepare a flexible budget with more than one cost driver.

LO9–6 Understand common errors made in preparing performance reports based on budgets and actual results.

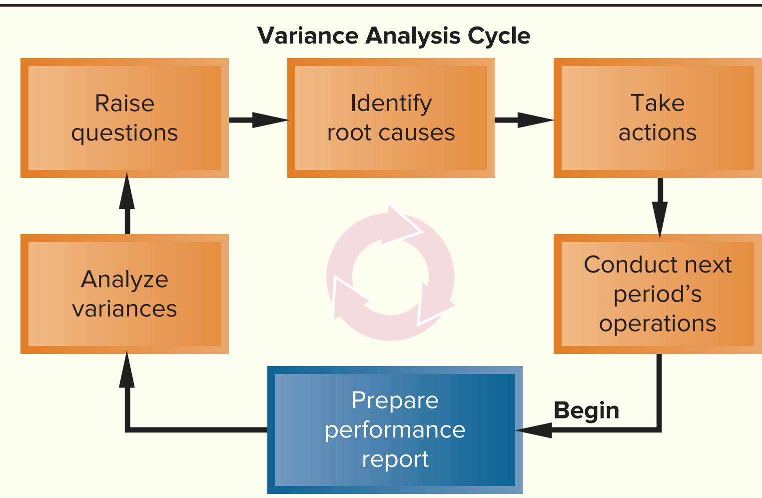
In the last chapter we explored how budgets are developed before a period begins. In this chapter, we explain how budgets can be adjusted to help guide actual operations and influence the performance evaluation process. For example, an organization's actual expenses will rarely equal its budgeted expenses as estimated at the beginning of the period. The reason is that the actual level of activity (such as unit sales) will rarely be the same as the budgeted activity; therefore, many actual expenses and revenues will naturally differ from what was budgeted. Should a manager be penalized for spending 10% more than budgeted for a variable expense like direct materials if unit sales are 10% higher than budgeted? Of course not. After studying this chapter, you'll know how to adjust a budget to enable meaningful comparisons to actual results.

The Variance Analysis Cycle

Companies use the *variance analysis cycle,* as illustrated in Exhibit 9–1, to evaluate and improve performance. The cycle begins with the preparation of performance reports in the accounting department. These reports highlight variances, which are the differences between the actual results and what should have occurred according to the budget. The variances raise questions. Why did this variance occur? Why is this variance larger than it was last period? The significant variances are investigated so that their root causes can be either replicated or eliminated. Then, next period's operations are carried out and the cycle begins again with the preparation of a new performance report for the latest period. The emphasis should be on highlighting superior and unsatisfactory results, finding the root causes of these outcomes, and then replicating the sources of superior achievement and eliminating the sources of unsatisfactory performance. The variance analysis cycle should not be used to assign blame for poor performance.

Managers frequently use the concept of *management by exception* in conjunction with the variance analysis cycle. **Management by exception** is a management system that compares actual results to a budget so that significant deviations can be flagged as exceptions and investigated further. This approach enables managers to focus on the most important variances while bypassing trivial discrepancies between the budget and actual results. For example, a variance of $5 is probably not big enough to warrant attention, whereas a variance of $5,000 might be worth tracking down. Another clue is the size of the variance relative to the amount of spending. A variance that is only 0.1% of spending on an item is probably caused by random factors. On the other hand, a variance of 10% of spending is much more likely to be a signal that something is wrong. In addition to

EXHIBIT 9–1
The Variance Analysis Cycle



watching for unusually large variances, the pattern of the variances should be monitored. For example, a run of steadily mounting variances should trigger an investigation even though none of the variances is large enough by itself to warrant investigation.

Next, we explain how organizations use flexible budgets to compare actual results to what should have occurred according to the budget.

Flexible Budgets

Characteristics of a Flexible Budget

LO9–1
Prepare a planning budget and a flexible budget and understand how they differ from one another.

The budgets that we explored in the last chapter were *planning budgets.* A **planning budget** is prepared before the period begins and is valid for only the planned level of activity. A static planning budget is suitable for planning but is inappropriate for evaluating how well costs are controlled. If the actual level of activity differs from what was planned, it would be misleading to compare actual costs to the static, unchanged planning budget. If activity is higher than expected, variable costs should be higher than expected; and if activity is lower than expected, variable costs should be lower than expected.

Flexible budgets take into account how changes in activity affect costs. A **flexible budget** is an estimate of what revenues and costs should have been, given the actual level of activity for the period. When a flexible budget is used in performance evaluation, actual costs are compared to what the costs *should have been for the actual level of activity during the period* rather than to the static planning budget. This is a very important distinction. If adjustments for the level of activity are not made, it is very difficult to interpret discrepancies between budgeted and actual costs.

IN BUSINESS

WINNERS AND LOSERS FROM THE NBA LOCKOUT

A company's actual net operating income can deviate from the budget for numerous and often uncontrollable reasons. For example, when the **National Basketball Association** (NBA) decided to suspend play because of a dispute between its team owners and players, many small businesses suffered—caterers, sports bars, apparel retailers, and parking lot owners all experienced a drop in revenues. **BestSportsApparel.com** experienced a substantial drop in NBA apparel sales due to the work stoppage. Rather than hiring 12 extra employees for the NBA season, the company reduced the size of its workforce.

While some companies lost revenues when the NBA shut down, others benefited from the situation. Andrew Zimbalist, professor of economics at Smith College, notes that "local economies are not impacted by sports work stoppages" because people choose to spend their entertainment dollars at other venues such as the theater, the zoo, or the museum.

Source: Emily Maltby and Sarah E. Needleman, "NBA Lockout: Local Firms Lose Big," *The Wall Street Journal,* October 13, 2011, p. B5.

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Deficiencies of the Static Planning Budget

To illustrate the difference between a static planning budget and a flexible budget, consider Rick's Hairstyling, an upscale hairstyling salon located in Beverly Hills that is owned and managed by Rick Manzi. Recently Rick has been attempting to get better control of his revenues and costs, and at the urging of his accounting and business adviser, Victoria Kho, he has begun to prepare monthly budgets.

At the end of February, Rick prepared the March budget that appears in Exhibit 9–2. Rick believes that the number of customers served in a month (also known as the number of client-visits) is the best way to measure the overall level of activity in his salon. A customer who comes into the salon and has his or her hair styled is counted as one client-visit.

EXHIBIT 9–2
Planning Budget

Rick's Hairstyling Planning Budget For the Month Ended March 31	
Budgeted client-visits *(q)*	1,000
Revenue ($180.00*q*)	$180,000
Expenses:	
Wages and salaries ($65,000 + $37.00*q*)	102,000
Hairstyling supplies ($1.50*q*)	1,500
Client gratuities ($4.10*q*)	4,100
Electricity ($1,500 + $0.10*q*)	1,600
Rent ($28,500)	28,500
Liability insurance ($2,800)	2,800
Employee health insurance ($21,300)	21,300
Miscellaneous ($1,200 + $0.20*q*)	1,400
Total expense	163,200
Net operating income	$ 16,800

Note that the term *revenue* is used in the planning budget rather than *sales.* We use the term revenue throughout the chapter because some organizations have sources of revenue other than sales. For example, donations, as well as sales, are counted as revenue in nonprofit organizations.

Rick has identified eight major categories of costs—wages and salaries, hairstyling supplies, client gratuities, electricity, rent, liability insurance, employee health insurance, and miscellaneous. Client gratuities consist of flowers, candies, and glasses of champagne that Rick gives to his customers while they are in the salon.

Working with Victoria, Rick estimated a cost formula for each cost. For example, the cost formula for electricity is $1,500 + $0.10*q*, where *q* equals the number of client-visits. In other words, electricity is a mixed cost with a $1,500 fixed element and a $0.10 per client-visit variable element. Once the budgeted level of activity was set at 1,000 client-visits, Rick computed the budgeted amount for each line item in the budget. For example, using the cost formula, he set the budgeted cost for electricity at $1,600 (= $1,500 + $0.10 × 1,000). To finalize his budget, Rick computed his expected net operating income for March of $16,800.

At the end of March, Rick prepared the income statement in Exhibit 9–3, which shows that 1,100 clients actually visited his salon in March and that his actual net operating income for the month was $21,230. It is important to realize that the actual results are *not* determined by plugging the actual number of client-visits into the revenue and cost formulas. The formulas are simply estimates of what the revenues and costs should be for a given level of activity. What actually happens usually differs from what is supposed to happen.

The first thing Rick noticed when comparing Exhibits 9–2 and 9–3 is that the actual profit of $21,230 (from Exhibit 9–3) was substantially higher than the budgeted profit of $16,800 (from Exhibit 9–2). This was, of course, good news, but Rick wanted to know more. Business was up by 10%—the salon had 1,100 client-visits instead of the budgeted 1,000 client-visits. Could this alone explain the higher net operating income? The answer is no. An increase in net operating income of 10% would have resulted in net operating income of only $18,480 (= 1.1 × $16,800), not the $21,230 actually earned during the month. What is responsible for this better outcome? Higher prices? Lower costs? Something else? Whatever the cause, Rick would like to know the answer and then hopefully repeat the same performance next month.

In an attempt to analyze what happened in March, Rick prepared the report comparing actual to budgeted costs that appears in Exhibit 9–4. Note that most of the variances in this report are labeled unfavorable (U) rather than favorable (F) even though net

EXHIBIT 9–3
Actual Results—Income Statement

Rick's Hairstyling Income Statement For the Month Ended March 31	
Actual client-visits	1,100
Revenue	$194,200
Expenses:	
Wages and salaries	106,900
Hairstyling supplies	1,620
Client gratuities	6,870
Electricity	1,550
Rent	28,500
Liability insurance	2,800
Employee health insurance	22,600
Miscellaneous	2,130
Total expense	172,970
Net operating income	$ 21,230

EXHIBIT 9–4
Comparison of Actual Results to the Static Planning Budget

Rick's Hairstyling
Comparison of Actual Results to the Planning Budget
For the Month Ended March 31

	Actual Results	Planning Budget	Variances*
Client-visits	1,100	1,000	
Revenue	$194,200	$180,000	$14,200 F
Expenses:			
Wages and salaries	106,900	102,000	4,900 U
Hairstyling supplies	1,620	1,500	120 U
Client gratuities	6,870	4,100	2,770 U
Electricity	1,550	1,600	50 F
Rent	28,500	28,500	0
Liability insurance	2,800	2,800	0
Employee health insurance	22,600	21,300	1,300 U
Miscellaneous	2,130	1,400	730 U
Total expense	172,970	163,200	9,770 U
Net operating income	$ 21,230	$ 16,800	$ 4,430 F

*The revenue variance is labeled favorable (unfavorable) when the actual revenue is greater than (less than) the planning budget. The expense variances are labeled favorable (unfavorable) when the actual expense is less than (greater than) the planning budget.

operating income was actually higher than expected. For example, wages and salaries show an unfavorable variance of $4,900 because the actual wages and salaries expense was $106,900, whereas the budget called for wages and salaries of $102,000. The problem with the report, as Rick immediately realized, is that it compares revenues and costs at one level of activity (1,000 client-visits) to revenues and costs at a different level of activity (1,100 client-visits). This is like comparing apples to oranges. Because Rick had 100 more client-visits than expected, some of his costs should be higher than budgeted. From Rick's standpoint, the increase in activity was good; however, it appears to be having a negative impact on most of the costs in the report. Rick knew that something would have to be done to make the report more meaningful, but he was unsure of what to do. So he made an appointment to meet with Victoria Kho to discuss the next step.

MANAGERIAL ACCOUNTING IN ACTION
THE ISSUE

RICK'S Hairstyling Salon

Victoria: How is the budgeting going?
Rick: Pretty well. I didn't have any trouble putting together the budget for March. I also prepared a report comparing the actual results for March to the budget, but that report isn't giving me what I really want to know.
Victoria: Because your actual level of activity didn't match your budgeted activity?
Rick: Right. I know the level of activity shouldn't affect my fixed costs, but we had more client-visits than I had expected and that had to affect my other costs.
Victoria: So you want to know whether the higher actual costs are justified by the higher level of activity?
Rick: Precisely.
Victoria: If you leave your reports and data with me, I can work on it later today, and by tomorrow I'll have a report to show you.

How a Flexible Budget Works

A flexible budget adjusts to show what costs *should be* for the actual level of activity. To illustrate how flexible budgets work, Victoria prepared the report in Exhibit 9–5 that shows what the *revenues and costs should have been given the actual level of activity* in March. Preparing the report is straightforward. The cost formula for each cost is used to estimate what the cost should have been for 1,100 client-visits—the actual level of activity for March. For example, using the cost formula $1,500 + $0.10*q*, the cost of electricity in March *should have been* $1,610 (= $1,500 + $0.10 × 1,100). Also, notice that the amounts of rent ($28,500), liability insurance ($2,800), and employee health insurance ($21,300) in the flexible budget equal the corresponding amounts in the planning budget (see Exhibit 9–2). This occurs because fixed costs are not affected by the activity level.

We can see from the flexible budget that the net operating income in March *should have been* $30,510, but recall from Exhibit 9–3 that the net operating income was actually only $21,230. The results are not as good as we thought. Why? We will answer that question shortly.

To summarize to this point, Rick had budgeted for a profit of $16,800. The actual profit was quite a bit higher—$21,230. However, Victoria's analysis shows that given the

EXHIBIT 9–5
Flexible Budget Based on Actual Activity

Rick's Hairstyling
Flexible Budget
For the Month Ended March 31

Actual client-visits (*q*)	1,100
Revenue ($180.000)*q*	$198,000
Expenses:	
Wages and salaries ($65,000 + $37.00*q*)	105,700
Hairstyling supplies ($1.50*q*)	1,650
Client gratuities ($4.10*q*)	4,510
Electricity ($1,500 + $0.10*q*)	1,610
Rent ($28,500)	28,500
Liability insurance ($2,800)	2,800
Employee health insurance ($21,300)	21,300
Miscellaneous ($1,200 + $0.20*q*)	1,420
Total expense	167,490
Net operating income	$ 30,510

actual number of client-visits in March, the profit should have been even higher—$30,510. What are the causes of these discrepancies? Rick would certainly like to build on the positive factors, while working to reduce the negative factors. But what are they?

Flexible Budget Variances

To answer Rick's questions concerning the discrepancies between budgeted and actual costs, Victoria broke down the variances shown in Exhibit 9–4 into two types of variances—activity variances and revenue and spending variances. We explain how she did it in the next two sections.

Activity Variances

LO9–2
Calculate and interpret activity variances.

Part of the discrepancy between the budgeted profit and the actual profit is due to the fact that the actual level of activity in March was higher than expected. How much of this discrepancy was due to this single factor? Victoria prepared the report in Exhibit 9–6 to answer this question. In that report, the flexible budget based on the actual level of activity for the period is compared to the planning budget from the beginning of the period. The flexible budget shows what should have happened at the actual level of activity, whereas the planning budget shows what should have happened at the budgeted level of activity. Therefore, the differences between the flexible budget and the planning budget show what should have happened solely because the actual level of activity differed from what had been expected.

For example, the flexible budget based on 1,100 client-visits shows revenue of $198,000 (= $180 per client-visit × 1,100 client-visits). The planning budget based on 1,000 client-visits shows revenue of $180,000 (= $180 per client-visit × 1,000 client-visits). Because the salon had 100 more client-visits than anticipated in the

EXHIBIT 9–6
Activity Variances from Comparing the Flexible Budget Based on Actual Activity to the Planning Budget

Rick's Hairstyling
Activity Variances
For the Month Ended March 31

	Flexible Budget	Planning Budget	Activity Variances*
Client-visits	1,100	1,000	
Revenue ($180.00*q*)	$198,000	$180,000	$18,000 F
Expenses:			
Wages and salaries ($65,000 + $37.00*q*)	105,700	102,000	3,700 U
Hairstyling supplies ($1.50*q*)	1,650	1,500	150 U
Client gratuities ($4.10*q*)	4,510	4,100	410 U
Electricity ($1,500 + $0.10*q*)	1,610	1,600	10 U
Rent ($28,500)	28,500	28,500	0
Liability insurance ($2,800)	2,800	2,800	0
Employee health insurance ($21,300)	21,300	21,300	0
Miscellaneous ($1,200 + $0.20*q*)	1,420	1,400	20 U
Total expense	167,490	163,200	4,290 U
Net operating income	$ 30,510	$ 16,800	$13,710 F

*The revenue variance is labeled favorable (unfavorable) when the revenue in the flexible budget is greater than (less than) the planning budget. The expense variances are labeled favorable (unfavorable) when the expense in the flexible budget is less than (greater than) the planning budget.

planning budget, actual revenue should have been higher than planned revenue by \$18,000 (= \$198,000 − \$180,000). This activity variance is shown on the report as \$18,000 F (favorable). Similarly, the flexible budget based on 1,100 client-visits shows electricity cost of \$1,610 (= \$1,500 + \$0.10 per client-visit × 1,100 client-visits). The planning budget based on 1,000 client-visits shows electricity cost of \$1,600 (= \$1,500 + \$0.10 per client-visit × 1,000 client-visits). Because the salon had 100 more client-visits than anticipated in the planning budget, the actual electricity cost should have been higher than the planned cost by \$10 (= \$1,610 − \$1,600). The activity variance for electricity is shown on the report as \$10 U (unfavorable). Note that in this case, the label "unfavorable" may be a little misleading. The electricity cost *should* be \$10 higher because business was up by 100 client-visits; therefore, it would be misleading to describe this variance in negative terms given that it was a necessary cost of serving more customers. For reasons such as this, we would like to caution you against assuming that unfavorable variances always indicate bad performance and favorable variances always indicate good performance.

Because all of the variances on this report are solely due to the difference between the actual level of activity and the level of activity in the planning budget from the beginning of the period, they are called **activity variances**. For example, the activity variance for revenue is \$18,000 F, the activity variance for electricity is \$10 U, and so on. The most important activity variance appears at the very bottom of the report; namely, the \$13,710 F (favorable) variance for net operating income. This variance says that because activity was higher than expected in the planning budget, the net operating income should have been \$13,710 higher. We caution against placing too much emphasis on any other single variance in this report. As we have said above, one would expect some costs to be higher as a consequence of more business. It is misleading to think of these unfavorable variances as indicative of poor performance.

On the other hand, the favorable activity variance for net operating income is important. Let's explore this variance a bit more thoroughly. First, as we have already noted, activity was up by 10%, but the flexible budget indicates that net operating income should have increased much more than 10%. A 10% increase in net operating income from the \$16,800 in the planning budget would result in net operating income of \$18,480 (= 1.1 × \$16,800); however, the flexible budget shows much higher net operating income of \$30,510. Why? The short answer is: Because of the presence of fixed costs. When we apply the 10% increase to the budgeted net operating income to estimate the profit at the higher level of activity, we implicitly assume that the revenues and *all* of the costs increase by 10%. But they do not. Note that when the activity level increases by 10%, three of the costs—rent, liability insurance, and employee health insurance—do not increase at all. These are all entirely fixed costs. So while sales do increase by 10%, these costs do not increase. This results in net operating income increasing by more than 10%. A similar effect occurs with the mixed costs, which contain fixed cost elements—wages and salaries, electricity, and miscellaneous. While sales increase by 10%, these mixed costs increase by less than 10%, resulting in an overall increase in net operating income of more than 10%. Because of the existence of fixed costs, net operating income does not change in proportion to changes in the level of activity. There is a leverage effect. The percentage changes in net operating income are ordinarily larger than the percentage increases in activity.

Revenue and Spending Variances

LO9–3
Calculate and interpret revenue and spending variances.

In the last section we answered the question "What impact did the change in activity have on our revenues, costs, and profit?" In this section we will answer the question "How well did we control our revenues, our costs, and our profit?"

Recall that the flexible budget based on the actual level of activity in Exhibit 9–5 shows what *should have happened given the actual level of activity*. Therefore, Victoria's next step was to compare actual results to the flexible budget—in essence comparing what actually happened to what should have happened. Her work is shown in Exhibit 9–7.

EXHIBIT 9–7
Revenue and Spending Variances from Comparing Actual Results to the Flexible Budget

Rick's Hairstyling
Revenue and Spending Variances
For the Month Ended March 31

	Actual Results	Flexible Budget	Revenue and Spending Variances*
Client-visits	1,100	1,100	
Revenue(180.00q$)	$194,200	$198,000	$3,800 U
Expenses:			
Wages and salaries ($65,000 + 37.00q$)	106,900	105,700	1,200 U
Hairstyling supplies (1.50q$)	1,620	1,650	30 F
Client gratuities (4.10q$)	6,870	4,510	2,360 U
Electricity ($1,500 + 0.10q$)	1,550	1,610	60 F
Rent ($28,500)	28,500	28,500	0
Liability insurance ($2,800)	2,800	2,800	0
Employee health insurance ($21,300)	22,600	21,300	1,300 U
Miscellaneous ($1,200 + 0.20q$)	2,130	1,420	710 U
Total expense	172,970	167,490	5,480 U
Net operating income	$ 21,230	$ 30,510	$9,280 U

*The revenue variance is labeled favorable (unfavorable) when the actual revenue is greater than (less than) the flexible budget. The expense variances are labeled favorable (unfavorable) when the actual expense is less than (greater than) the flexible budget.

Focusing first on revenue, the actual revenue totaled $194,200. However, the flexible budget indicates that, given the actual level of activity, revenue should have been $198,000. Consequently, revenue was $3,800 less than it should have been, given the actual number of client-visits for the month. This discrepancy is labeled as a $3,800 U (unfavorable) variance and is called a *revenue variance*. A **revenue variance** is the difference between the actual total revenue and what the total revenue should have been, given the actual level of activity for the period. If actual revenue exceeds what the revenue should have been, the variance is labeled favorable. If actual revenue is less than what the revenue should have been, the variance is labeled unfavorable. Why would actual revenue be less than or more than it should have been, given the actual level of

IN BUSINESS

THE SALES IMPLICATIONS OF THE WORLD CUP

Actual sales volumes can differ from planned sales volumes for many reasons. For example, some companies may fail to consider how sporting events such as the World Cup will affect their planned sales. **Corning** said TV sales were up 13% in Europe and 64% in Latin America during the World Cup and **Carrefour** reported a bump in sales of its beer, soft drinks, and meats during the same period. Conversely, **Whirlpool** reported lower demand for its washing machines (because the company claimed that customers were spending their disposable income on TVs) and Denny's reported a drop in full-service dining during the World Cup.

Source: Vipal Monga and Emily Chasan, "When in Doubt, Blame It on the World Cup," *The Wall Street Journal,* July 31, 2014, pp. B1 and B4.


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activity? Basically, the revenue variance is favorable if the average selling price is greater than expected; it is unfavorable if the average selling price is less than expected. This could happen for a variety of reasons including a change in selling price, a different mix of products sold, a change in the amount of discounts given, poor accounting controls, and so on.

Focusing next on costs, the actual electricity cost was $1,550; however, the flexible budget indicates that electricity costs should have been $1,610 for the 1,100 client-visits in March. Because the cost was $60 less than we would have expected for the actual level of activity during the period, it is labeled as a favorable variance, $60 F. This is an example of a *spending variance.* A **spending variance** is the difference between the actual amount of the cost and how much a cost should have been, given the actual level of activity. If the actual cost is greater than what the cost should have been, the variance is labeled as unfavorable. If the actual cost is less than what the cost should have been, the variance is labeled as favorable. Why would a cost have a favorable or unfavorable variance? There are many possible explanations including paying a higher price for inputs than should have been paid, using too many inputs for the actual level of activity, a change in technology, and so on. In the next chapter we will explore these types of explanations in greater detail.

Note from Exhibit 9–7 that the overall net operating income variance is $9,280 U (unfavorable). This means that given the actual level of activity for the period, the net operating income was $9,280 lower than it should have been. There are a number of reasons for this. The most prominent is the unfavorable revenue variance of $3,800. Next in line is the $2,360 unfavorable variance for client gratuities. Looking at this in another way, client gratuities were more than 50% larger than they should have been according to the flexible budget. This is a variance that Rick would almost certainly want to investigate further. He may find that this unfavorable variance is not necessarily a bad thing. It is possible, for example, that more lavish use of gratuities led to the 10% increase in client-visits.

Exhibit 9–7 also includes a $1,300 unfavorable variance related to employee health insurance, thereby highlighting how a fixed cost can have a spending variance. While fixed costs do not depend on the level of activity, the actual amount of a fixed cost can differ from the estimated amount included in a flexible budget. For example, perhaps Rick's employee health insurance premiums unexpectedly increased by $1,300 during March.

In conclusion, the revenue and spending variances in Exhibit 9–7 will help Rick better understand why his actual net operating income differs from what should have happened given the actual level of activity.

A Performance Report Combining Activity and Revenue and Spending Variances

LO9–4
Prepare a performance report that combines activity variances and revenue and spending variances.

Exhibit 9–8 displays Victoria's performance report that combines the activity variances (from Exhibit 9–6) with the revenue and spending variances (from Exhibit 9–7). The report brings together information from those two earlier exhibits in a way that makes it easier to interpret what happened during the period. The format of this report is a bit different from the format of the previous reports in that the variances appear between the amounts being compared rather than after them. For example, the activity variances appear between the flexible budget amounts and the planning budget amounts. In Exhibit 9–6, the activity variances appeared after the flexible budget and the planning budget.

Note two numbers in particular in the performance report—the activity variance for net operating income of $13,710 F (favorable) and the overall revenue and spending variance for net operating income of $9,280 U (unfavorable). It is worth repeating what those two numbers mean. The $13,710 favorable activity variance occurred because actual activity (1,100 client-visits) was greater than the budgeted level of activity

EXHIBIT 9–8
Performance Report Combining Activity Variances with Revenue and Spending Variances

Rick's Hairstyling
Flexible Budget Performance Report
For the Month Ended March 31

	(1) Actual Results	Revenue and Spending Variances (1) – (2)	(2) Flexible Budget	Activity Variances (2) – (3)	(3) Planning Budget
Client-visits	1,100		1,100		1,000
Revenue (180.00q$)	$194,200	$3,800 U	$198,000	$18,000 F	$180,000
Expenses:					
Wages and salaries ($65,000 + 37.00q$)	106,900	1,200 U	105,700	3,700 U	102,000
Hairstyling supplies (1.50q$)	1,620	30 F	1,650	150 U	1,500
Client gratuities (4.10q$)	6,870	2,360 U	4,510	410 U	4,100
Electricity ($1,500 + 0.10q$)	1,550	60 F	1,610	10 U	1,600
Rent ($28,500)	28,500	0	28,500	0	28,500
Liability insurance ($2,800)	2,800	0	2,800	0	2,800
Employee health insurance ($21,300)	22,600	1,300 U	21,300	0	21,300
Miscellaneous ($1,200 + 0.20q$)	2,130	710 U	1,420	20 U	1,400
Total expense	172,970	5,480 U	167,490	4,290 U	163,200
Net operating income	$ 21,230	$9,280 U	$ 30,510	$13,710 F	$ 16,800

(1,000 client-visits). The $9,280 unfavorable overall revenue and spending variance occurred because the profit was not as large as it should have been for the actual level of activity for the period. These two different variances mean very different things and call for different types of actions. To generate a favorable activity variance for net operating income, managers must take actions to increase client-visits. To generate a favorable overall revenue and spending variance, managers must take actions to protect selling prices, increase operating efficiency, and reduce the prices of inputs.

The performance report in Exhibit 9–8 provides much more useful information to managers than a simple comparison of the planning budget with actual results as shown in Exhibit 9–4. In Exhibit 9–4, the effects of changes in activity were jumbled together with the effects of how well prices were controlled and operations were managed. The performance report in Exhibit 9–8 clearly separates these effects, allowing managers to take a much more focused approach in evaluating operations.

To get a better idea of how the performance report accomplishes this task, look at hairstyling supplies in the performance report. The actual cost of hairstyling supplies was $1,620 for the period, whereas in the planning budget, this cost was $1,500. In the comparison of the actual results to the planning budget in Exhibit 9–4, this difference is shown as an unfavorable variance of $120. Exhibit 9–4 uses a static planning budget approach that compares actual costs at one level of activity to budgeted costs at a different level of activity. As we said before, this is like comparing apples to oranges. This variance is actually a mixture of two different effects. This becomes clear in the performance report in Exhibit 9–8. The $120 difference between the actual results and the budgeted amount is composed of two different variances—a favorable spending variance of $30 and an unfavorable activity variance of $150. The favorable spending variance occurred because less was spent on hairstyling supplies than one would have expected, given the actual level of activity for the month. The activity variance occurs because activity was greater than anticipated in the planning budget, which naturally resulted in a higher total cost for this variable cost.

The flexible budget performance report in Exhibit 9–8 provides a more valid assessment of performance than simply comparing actual costs to static planning budget costs because actual costs are compared to what costs should have been at the actual level of activity. In other words, apples are compared to apples. When this is done, we see that the spending variance for hairstyling supplies is $30 F (favorable) rather than $120 U (unfavorable) as it was in the original static planning budget performance report (see Exhibit 9–4). In some cases, as with hairstyling supplies in Rick's report, an unfavorable static planning budget variance may be transformed into a favorable revenue or spending variance when an increase in activity is properly taken into account. The following discussion took place the next day at Rick's salon.

MANAGERIAL ACCOUNTING IN ACTION
THE WRAP-UP

RICK'S
Hairstyling Salon

Victoria: Let me show you what I've got. [Victoria shows Rick the flexible budget performance report in Exhibit 9–8.] I simply used the cost formulas to update the budget to reflect the increase in client-visits you experienced in March. That allowed me to come up with a better benchmark for what the costs should have been.

Rick: That's what you labeled the "flexible budget based on 1,100 client-visits"?

Victoria: That's right. Your original budget was based on 1,000 client-visits, so it understated what some of the costs should have been when you actually served 1,100 customers.

Rick: That's clear enough. These spending variances aren't quite as shocking as the variances on my first report.

Victoria: Yes, but you still have an unfavorable variance of $2,360 for client gratuities.

Rick: I know how that happened. In March there was a big political fundraising dinner that I forgot about when I prepared the March budget. To fit all of our regular clients in, we had to push them through here pretty fast. Everyone still got top-rate service, but I felt bad about not being able to spend as much time with each customer. I wanted to give my customers a little extra something to compensate them for the less personal service, so I ordered a lot of flowers, which I gave away by the bunch.

Victoria: With the prices you charge, Rick, I am sure the gesture was appreciated.

Rick: One thing bothers me about the report. When we discussed my costs before, you called rent, liability insurance, and employee health insurance fixed costs. How can I have a variance for a fixed cost? Doesn't fixed mean that it doesn't change?

Victoria: We call these costs *fixed* because they shouldn't be affected by *changes in the level of activity*. However, that doesn't mean that they can't change for other reasons. Also, the use of the term *fixed* suggests to people that the cost can't be controlled, but that isn't true. It is often easier to control fixed costs than variable costs. For example, it would be fairly easy for you to change your insurance bill by adjusting the amount of insurance you carry. It would be much more difficult for you to significantly reduce your spending on hairstyling supplies—a variable cost that is a necessary part of serving customers.

Rick: I think I understand, but it *is* confusing.

Victoria: Just remember that a cost is called variable if it is proportional to activity; it is called fixed if it does not depend on the level of activity. However, fixed costs can change for reasons unrelated to changes in the level of activity. And controllability has little to do with whether a cost is variable or fixed. Fixed costs are often more controllable than variable costs.

IN BUSINESS

CHILLY SPRING WEATHER BRINGS BIG DISCOUNTS AND LOWER MARGINS

Cold temperatures can keep shoppers away from malls. When April temperatures dropped below expectations so did sales at **Victoria's Secret**, **Bath & Body Works**, and **Aeropostale**. Fewer shoppers also can lead to big discounts as Aeropostale offered 50% off shorts to those who did choose to shop in its stores.

When these types of retailers establish their budgets for the spring, it is impossible for them to foresee months in advance how the weather may influence their sales, which in turn can have an impact on their revenue activity variances.

Source: Anna Prior and Karen Talley, "Weather Holds Back Retailers," *The Wall Street Journal,* May 10, 2013, p. B2.

© Austin Bush/Getty Images

Performance Reports in Nonprofit Organizations

The performance reports in nonprofit organizations are basically the same as the performance reports we have considered so far—with one prominent difference. Nonprofit organizations usually receive a significant amount of funding from sources other than sales. For example, universities receive their funding from sales (i.e., tuition charged to students), from endowment income and donations, and—in the case of public universities—from state appropriations. This means that, like costs, the revenue in governmental and nonprofit organizations may consist of both fixed and variable elements. For example, the **Seattle Opera Company**'s revenue in a recent year consisted of grants and donations of $12,719,000 and ticket sales of $8,125,000 (or about $75.35 per ticket sold). Consequently, the revenue formula for the opera can be written as:

$$\text{Revenue} = \$12{,}719{,}000 + \$75.35q$$

where q is the number of tickets sold. In other respects, the performance report for the Seattle Opera and other nonprofit organizations would be similar to the performance report in Exhibit 9–8.

Performance Reports in Cost Centers

Performance reports are often prepared for organizations that do not have any source of outside revenue. In particular, in a large organization a performance report may be prepared for each department—including departments that do not sell anything to outsiders. For example, a performance report is very commonly prepared for production departments in manufacturing companies. Such reports should be prepared using the same principles we have discussed and should look very much like the performance report in Exhibit 9–8—with the exception that revenue, and consequently net operating income, will not appear on the report. Because the managers in these departments are responsible for costs, but not revenues, they are often called *cost centers.*

Flexible Budgets with Multiple Cost Drivers

LO9–5
Prepare a flexible budget with more than one cost driver.

At Rick's Hairstyling, we have thus far assumed that there is only one cost driver—the number of client-visits. However, in the activity-based costing chapter, we found that more than one cost driver might be needed to adequately explain all of the costs in an organization. For example, some of the costs at Rick's Hairstyling probably depend more on the number of hours that the salon is open for business than the number of client-visits. Specifically, most of Rick's employees are paid salaries, but some are paid on an hourly basis. None of the

EXHIBIT 9–9
Flexible Budget Based on More than One Cost Driver

Rick's Hairstyling
Flexible Budget
For the Month Ended March 31

Actual client-visits (q_1)	1,100
Actual hours of operation (q_2)	190
Revenue (180.00q_1$)	$198,000
Expenses:	
Wages and salaries ($65,000 + 220q_2$)	106,800
Hairstyling supplies (1.50q_1$)	1,650
Client gratuities (4.10q_1$)	4,510
Electricity ($390 + 0.10q_1$ + 6.00q_2$)	1,640
Rent ($28,500)	28,500
Liability insurance ($2,800)	2,800
Employee health insurance ($21,300)	21,300
Miscellaneous ($1,200 + 0.20q_1$)	1,420
Total expense	168,620
Net operating income	$ 29,380

employees is paid on the basis of the number of customers actually served. Consequently, the cost formula for wages and salaries would be more accurate if it were stated in terms of the hours of operation rather than the number of client-visits. The cost of electricity is even more complex. Some of the cost is fixed—the heat must be kept at some minimum level even at night when the salon is closed. Some of the cost depends on the number of client-visits—the power consumed by hair dryers depends on the number of customers served. Some of the cost depends on the number of hours the salon is open—the costs of lighting the salon and heating it to a comfortable temperature. Consequently, the cost formula for electricity would be more accurate if it were stated in terms of both the number of client-visits and the hours of operation rather than just in terms of the number of client-visits.

Exhibit 9–9 shows a flexible budget in which these changes have been made. In that flexible budget, two cost drivers are listed—client-visits and hours of operation—where q_1 refers to client-visits and q_2 refers to hours of operation. For example, wages and salaries depend on the hours of operation and its cost formula is $65,000 + 220q_2$. Because the salon actually operated 190 hours, the flexible budget amount for wages and salaries is $106,800 (= $65,000 + $220 × 190). The electricity cost depends on

IN BUSINESS

ON-CALL SCHEDULING DRAWS THE ATTENTION OF THE NEW YORK ATTORNEY GENERAL

The New York Attorney General has warned **Target, Gap**, and 11 other companies that their on-call employee scheduling practices may violate the law. These companies are using software programs to forecast immediate-term staffing needs based on real-time sales and customer traffic information. If a store is busy it requires its on-call employees to come to work, whereas if the store is not busy the employees do not come to work and they are not paid. In other words, the employees need to plan to be available even though they may or may not be called into work or get paid.

From a flexible budgeting standpoint, the companies are trying to make their labor costs variable with respect to sales and customer traffic. From a legal standpoint, the Attorney General noted that this compensation scheme leaves employees "too little time to make arrangements for family needs, let alone to find an alternative source of income to compensate for the lost pay."

Source: Lauren Weber, "Retailers Under Fire for Work Schedules," *The Wall Street Journal,* April 13, 2015, pp. B1–B2.

both client-visits and the hours of operation and its cost formula is $\$390 + \$0.10q_1 + \$6.00q_2$. Because the actual number of client-visits was 1,100 and the salon actually operated for 190 hours, the flexible budget amount for electricity is $1,640 (= $390 + $0.10 × 1,100 + $6.00 × 190). Notice that the net operating income in the flexible budget based on two cost drivers is $29,380, whereas the net operating income in the flexible budget based on one cost driver (see Exhibit 9–5) is $30,510. These two amounts differ because the flexible budget based on two cost drivers is more accurate than the flexible budget based on one driver.

The revised flexible budget based on both client-visits and hours of operation can be used exactly like we used the earlier flexible budget based on just client-visits to compute activity variances as in Exhibit 9–6, revenue and spending variances as in Exhibit 9–7, and a performance report as in Exhibit 9–8. The difference is that because the cost formulas based on more than one cost driver are more accurate than the cost formulas based on just one cost driver, the variances will also be more accurate.

Some Common Errors

LO9–6
Understand common errors made in preparing performance reports based on budgets and actual results.

We started this chapter by discussing the need for managers to understand the difference between what actually happened and what was expected to happen—formalized by the planning budget. To meet this need, we developed a flexible budget that allowed us to isolate activity variances and revenue and spending variances. Unfortunately, this approach is not always followed in practice—resulting in misleading and difficult-to-interpret reports. The most common errors in preparing performance reports are to implicitly assume that all costs are fixed or to implicitly assume that all costs are variable. These erroneous assumptions lead to inaccurate benchmarks and incorrect variances.

We have already discussed one of these errors—assuming that all costs are fixed. This is the error that is made when static planning budget costs are compared to actual costs without any adjustment for the actual level of activity. Such a comparison appeared in Exhibit 9–4. For convenience, the comparison of actual to planned revenues and costs is repeated in Exhibit 9–10. Looking at that exhibit, note that the actual cost of hairstyling supplies of $1,620 is directly compared to the planned cost of $1,500, resulting in an unfavorable variance of $120. But this comparison only makes sense if the cost of hairstyling supplies is fixed. If the cost of hairstyling supplies isn't fixed (and indeed it is not), one would *expect*

EXHIBIT 9–10
Faulty Analysis Comparing Actual Amounts to Planned Amounts (Implicitly Assumes All Income Statement Items Are Fixed)

Rick's Hairstyling
For the Month Ended March 31

	Actual Results	Planning Budget	Variances
Client-visits	1,100	1,000	
Revenue	$194,200	$180,000	$14,200 F
Expenses:			
Wages and salaries	106,900	102,000	4,900 U
Hairstyling supplies	1,620	1,500	120 U
Client gratuities	6,870	4,100	2,770 U
Electricity	1,550	1,600	50 F
Rent	28,500	28,500	0
Liability insurance	2,800	2,800	0
Employee health insurance	22,600	21,300	1,300 U
Miscellaneous	2,130	1,400	730 U
Total expense	172,970	163,200	9,770 U
Net operating income	$ 21,230	$ 16,800	$ 4,430 F

IN BUSINESS

© Getty Images RF

SNOW REMOVAL BUDGETS GET PLOWED UNDER

The **District of Columbia** budgeted $6.2 million for its annual snow removal needs based on an expectation of 15 inches of snow per year. So, when it received 28 inches of snow in one weekend, its snow removal budget got plowed under. During this same weekend, the **Virginia Department of Transportation** estimated that 500,000 tons of snow dropped on its northern Virginia roadways, leading to historic cost overruns in its $79 million snow removal budget.

In these types of situations, a flexible budget can help managers assess operational efficiency. For example, a manager could create a cost formula such as "the snow removal cost per inch of snow" to estimate what the total variable snow removal costs should be for the actual inches of snowfall. This "flexed" snow removal cost could be compared to the actual snow removal cost in an effort to identify sources of exceptional performance or opportunities for process improvement.

Source: Sudeep Reddy and Clare Ansberry, "States Face Big Costs to Dig Out From Blizzard," *The Wall Street Journal,* February 9, 2010, p. A6.

the cost to go up because of the increase in activity over the planning budget. Comparing actual costs to static planning budget costs only makes sense if the cost is fixed. If the cost isn't fixed, it needs to be adjusted for any change in activity that occurs during the period.

The other common error when comparing a planning budget to actual results is to assume that all costs are variable. A report that makes this error appears in Exhibit 9–11. The variances in this report are computed by comparing actual results to the amounts in the second numerical column where *all* of the items in the planning budget have been inflated by 10%—the percentage by which activity increased. This is a perfectly valid adjustment to make if an item is strictly variable—like sales and hairstyling supplies. It is *not* a valid adjustment if the item contains any fixed element. Take, for example, rent. If the salon serves 10% more customers in a given month, would you expect the rent to increase by 10%? The answer is no. Ordinarily, the rent is fixed in advance and does not depend on the volume of business. Therefore, the amount shown in the second numerical column of $31,350 is incorrect, which leads to the erroneous favorable variance of $2,850. In fact, the actual rent paid was exactly equal to the budgeted rent, so there should be no variance at all on a valid report.

EXHIBIT 9–11
Faulty Analysis That Assumes All Items in the Planning Budget Are Variable

Rick's Hairstyling
For the Month Ended March 31

	(1) Actual Results	(2) Planning Budget × (1,100/1,000)	Variances (1) – (2)
Client-visits	1,100		
Revenue	$194,200	$198,000	$3,800 U
Expenses:			
Wages and salaries	106,900	112,200	5,300 F
Hairstyling supplies	1,620	1,650	30 F
Client gratuities	6,870	4,510	2,360 U
Electricity	1,550	1,760	210 F
Rent	28,500	31,350	2,850 F
Liability insurance	2,800	3,080	280 F
Employee health insurance	22,600	23,430	830 F
Miscellaneous	2,130	1,540	590 U
Total expense	172,970	179,520	6,550 F
Net operating income	$ 21,230	$ 18,480	$2,750 F

Summary

Directly comparing actual revenues and costs to static planning budget revenues and costs can easily lead to erroneous conclusions. Actual revenues and costs differ from budgeted revenues and costs for a variety of reasons, but one of the biggest is a change in the level of activity. One would expect actual revenues and costs to increase or decrease as the activity level increases or decreases. Flexible budgets enable managers to isolate the various causes of the differences between budgeted and actual costs.

A flexible budget is a budget that is adjusted to the actual level of activity. It is the best estimate of what revenues and costs should have been, given the actual level of activity during the period. The flexible budget can be compared to the budget from the beginning of the period or to the actual results.

When the flexible budget is compared to the planning budget, activity variances are the result. An activity variance shows how a revenue or cost should have changed in response to the difference between actual and planned activity.

When actual results are compared to the flexible budget, revenue and spending variances are the result. A favorable revenue variance indicates that revenue was larger than should have been expected, given the actual level of activity. An unfavorable revenue variance indicates that revenue was less than it should have been, given the actual level of activity. A favorable spending variance indicates that the cost was less than expected, given the actual level of activity. An unfavorable spending variance indicates that the cost was greater than it should have been, given the actual level of activity.

A flexible budget performance report combines activity variances and revenue and spending variances on one report.

Common errors in comparing actual costs to budgeted costs are to assume all costs are fixed or to assume all costs are variable. If all costs are assumed to be fixed, the variances for variable and mixed costs will be incorrect. If all costs are assumed to be variable, the variances for fixed and mixed costs will be incorrect. The variance for a cost will only be correct if the actual behavior of the cost is used to develop the flexible budget benchmark.

Review Problem: Variance Analysis Using a Flexible Budget

Harrald's Fish House is a family-owned restaurant that specializes in Scandinavian-style seafood. Data concerning the restaurant's monthly revenues and costs appear below (*q* refers to the number of meals served):

	Formula
Revenue	$16.50*q*
Cost of ingredients	$6.25*q*
Wages and salaries	$10,400
Utilities	$800 + $0.20*q*
Rent	$2,200
Miscellaneous	$600 + $0.80*q*

Required:

1. Prepare the restaurant's planning budget for April assuming that 1,800 meals are served.
2. Assume that 1,700 meals were actually served in April. Prepare a flexible budget for this level of activity.
3. The actual results for April appear below. Prepare a flexible budget performance report for the restaurant for April.

Revenue	$27,920
Cost of ingredients	$11,110
Wages and salaries	$10,130
Utilities	$1,080
Rent	$2,200
Miscellaneous	$2,240

Solution to Review Problem

1. The planning budget for April appears below:

Harrald's Fish House Planning Budget For the Month Ended April 30	
Budgeted meals served *(q)*	1,800
Revenue ($16.50*q*)	$29,700
Expenses:	
Cost of ingredients ($6.25*q*)	11,250
Wages and salaries ($10,400)	10,400
Utilities ($800 + $0.20*q*)	1,160
Rent ($2,200)	2,200
Miscellaneous ($600 + $0.80*q*)	2,040
Total expense	27,050
Net operating income	$ 2,650

2. The flexible budget for April appears below:

Harrald's Fish House Flexible Budget For the Month Ended April 30	
Actual meals served *(q)*	1,700
Revenue ($16.50*q*)	$28,050
Expenses:	
Cost of ingredients ($6.25*q*)	10,625
Wages and salaries ($10,400)	10,400
Utilities ($800 + $0.20*q*)	1,140
Rent ($2,200)	2,200
Miscellaneous ($600 + $0.80*q*)	1,960
Total expense	26,325
Net operating income	$ 1,725

3. The flexible budget performance report for April appears below:

Harrald's Fish House
Flexible Budget Performance Report
For the Month Ended April 30

	(1) Actual Results	Revenue and Spending Variances (1) – (2)	(2) Flexible Budget	Activity Variances (2) – (3)	(3) Planning Budget
Meals served	1,700		1,700		1,800
Revenue ($16.50*q*)	$27,920	$130 U	$28,050	$1,650 U	$29,700
Expenses:					
Cost of ingredients ($6.25*q*)	11,110	485 U	10,625	625 F	11,250
Wages and salaries ($10,400)	10,130	270 F	10,400	0	10,400
Utilities ($800 + $0.20*q*)	1,080	60 F	1,140	20 F	1,160
Rent ($2,200)	2,200	0	2,200	0	2,200
Miscellaneous ($600 + $0.80*q*)	2,240	280 U	1,960	80 F	2,040
Total expense	26,760	435 U	26,325	725 F	27,050
Net operating income	$ 1,160	$565 U	$ 1,725	$ 925 U	$ 2,650

Glossary

Activity variance The difference between a revenue or cost item in the flexible budget and the same item in the static planning budget. An activity variance is due solely to the difference between the actual level of activity used in the flexible budget and the level of activity assumed in the planning budget. (p. 420)

Flexible budget A report showing estimates of what revenues and costs should have been, given the actual level of activity for the period. (p. 415)

Management by exception A management system in which actual results are compared to a budget. Significant deviations from the budget are flagged as exceptions and investigated further. (p. 414)

Planning budget A budget created at the beginning of the budgeting period that is valid only for the planned level of activity. (p. 415)

Revenue variance The difference between the actual revenue for the period and how much the revenue should have been, given the actual level of activity. A favorable (unfavorable) revenue variance occurs because the revenue is higher (lower) than expected, given the actual level of activity for the period. (p. 421)

Spending variance The difference between the actual amount of the cost and how much the cost should have been, given the actual level of activity. A favorable (unfavorable) spending variance occurs because the cost is lower (higher) than expected, given the actual level of activity for the period. (p. 422)

Questions

9–1 What is a static planning budget?

9–2 What is a flexible budget and how does it differ from a static planning budget?

9–3 What are some of the possible reasons that actual results may differ from what had been budgeted at the beginning of a period?

9–4 Why is it difficult to interpret a difference between how much expense was budgeted and how much was actually spent?

9–5 What is an activity variance and what does it mean?

9–6 What is a revenue variance and what does it mean?

9–7 What is a spending variance and what does it mean?

9–8 What does a flexible budget performance report do that a simple comparison of budgeted to actual results does not do?

9–9 How does a flexible budget based on two cost drivers differ from a flexible budget based on one cost driver?

9–10 What assumption is implicitly made about cost behavior when actual results are directly compared to a static planning budget? Why is this assumption questionable?

9–11 What assumption is implicitly made about cost behavior when all of the items in a static planning budget are adjusted in proportion to a change in activity? Why is this assumption questionable?

Applying Excel

LO9–1, LO9–2, LO9–3, LO9–4

The Excel worksheet form that appears below is to be used to recreate the Review Problem relating to Harrald's Fish House. Download the workbook containing this form from Connect, where you will also receive instructions about how to use this worksheet form.

You should proceed to the requirements below only after completing your worksheet.

Required:

1. Check your worksheet by changing the revenue in cell D4 to \$16.00; the cost of ingredients in cell D5 to \$6.50; and the wages and salaries in cell B6 to \$10,000. The activity variance for net operating income should now be \$850 U and the spending variance for total expenses should be \$410 U. If you do not get these answers, find the errors in your worksheet and correct them.
   a. What is the activity variance for revenue? Explain this variance.
   b. What is the spending variance for the cost of ingredients? Explain this variance.

	A	B	C	D	E	F	G	H
1	Chapter 9: Applying Excel							
2								
3	**Data**							
4	Revenue			$16.50 q				
5	Cost of ingredients			$6.25 q				
6	Wages and salaries	$10,400						
7	Utilities	$800	+	$0.20 q				
8	Rent	$2,200						
9	Miscellaneous	$600	+	$0.80 q				
10								
11	Actual results:							
12	Revenue	$27,920						
13	Cost of ingredients	$11,110						
14	Wages and salaries	$10,130						
15	Utilities	$1,080						
16	Rent	$2,200						
17	Miscellaneous	$2,240						
18								
19	Planning budget activity	1,800	meals served					
20	Actual activity	1,700	meals served					
21								
22	*Enter a formula into each of the cells marked with a ? below*							
23	**Review Problem: Variance Analysis Using a Flexible Budget**							
24								
25	***Construct a flexible budget performance report***							
26			*Revenue*					
27			*and*					
28		*Actual*	*Spending*		*Flexible*	*Activity*		*Planning*
29		*Results*	*Variances*		*Budget*	*Variances*		*Budget*
30	Meals served	?			?			?
31	Revenue	?	?		?	?		?
32	Expenses:							
33	Cost of ingredients	?	?		?	?		?
34	Wages and salaries	?	?		?	?		?
35	Utilities	?	?		?	?		?
36	Rent	?	?		?	?		?
37	Miscellaneous	?	?		?	?		?
38	Total expenses	?	?		?	?		?
39	Net operating income	?	?		?	?		?

Chapter 9 Form / Filled in Chapter 9 Form / Chapter 9 Form

2. Revise the data in your worksheet to reflect the results for the following year:

Data			
Revenue			16.50q$
Cost of ingredients			6.25q$
Wages and salaries	$10,400		
Utilities	$800	+	0.20q$
Rent	$2,200		
Miscellaneous	$600	+	0.80q$
Actual results:			
Revenue	$28,900		
Cost of ingredients	$11,300		
Wages and salaries	$10,300		
Utilities	$1,120		
Rent	$2,300		
Miscellaneous	$2,020		
Planning budget activity	1,700 meals served		
Actual activity	1,800 meals served		

Using the flexible budget performance report, briefly evaluate the company's performance for the year and indicate where attention should be focused.

connect The Foundational 15

LO9–1, LO9–2, LO9–3

Adger Corporation is a service company that measures its output based on the number of customers served. The company provided the following fixed and variable cost estimates that it uses for budgeting purposes and the actual results for May as shown below:

	Fixed Element per Month	Variable Element per Customer Served	Actual Total for May
Revenue		$5,000	$160,000
Employee salaries and wages	$50,000	$1,100	$88,000
Travel expenses		$600	$19,000
Other expenses	$36,000		$34,500

When preparing its planning budget the company estimated that it would serve 30 customers per month; however, during May the company actually served 35 customers.

Required *(all computations pertain to the month of May)*:

1. What amount of revenue would be included in Adger's flexible budget?
2. What amount of employee salaries and wages would be included in Adger's flexible budget?
3. What amount of travel expenses would be included in Adger's flexible budget?
4. What amount of other expenses would be included in Adger's flexible budget?
5. What net operating income would appear in Adger's flexible budget?
6. What is Adger's revenue variance?
7. What is Adger's employee salaries and wages spending variance?
8. What is Adger's travel expenses spending variance?
9. What is Adger's other expenses spending variance?
10. What amount of revenue would be included in Adger's planning budget?
11. What amount of employee salaries and wages would be included in Adger's planning budget?
12. What amount of travel expenses would be included in Adger's planning budget?
13. What amount of other expenses would be included in Adger's planning budget?
14. What activity variance would Adger report with respect to its revenue?
15. What activity variances would Adger report with respect to each of its expenses?

connect Exercises

EXERCISE 9–1 Prepare a Flexible Budget LO9–1

Puget Sound Divers is a company that provides diving services such as underwater ship repairs to clients in the Puget Sound area. The company's planning budget for May appears below:

Puget Sound Divers Planning Budget For the Month Ended May 31	
Budgeted diving-hours (*q*)	100
Revenue ($365.00*q*)	$36,500
Expenses:	
Wages and salaries ($8,000 + $125.00*q*)	20,500
Supplies ($3.00*q*)	300
Equipment rental ($1,800 + $32.00*q*)	5,000
Insurance ($3,400)	3,400
Miscellaneous ($630 + $1.80*q*)	810
Total expense	30,010
Net operating income	$ 6,490

During May, the company's actual activity was 105 diving-hours.

Required:

Using Exhibit 9–5 as your guide, prepare a flexible budget for May.

EXERCISE 9–2 Activity Variances LO9–2

Flight Café prepares in-flight meals for airlines in its kitchen located next to a local airport. The company's planning budget for July appears below:

Flight Café
Planning Budget
For the Month Ended July 31

Budgeted meals (*q*)	18,000
Revenue ($4.50*q*)	$81,000
Expenses:	
Raw materials ($2.40*q*)	43,200
Wages and salaries ($5,200 + $0.30*q*)	10,600
Utilities ($2,400 + $0.05*q*)	3,300
Facility rent ($4,300)	4,300
Insurance ($2,300)	2,300
Miscellaneous ($680 + $0.10*q*)	2,480
Total expense	66,180
Net operating income	$14,820

In July, 17,800 meals were actually served. The company's flexible budget for this level of activity appears below:

Flight Café
Flexible Budget
For the Month Ended July 31

Budgeted meals (*q*)	17,800
Revenue ($4.50*q*)	$80,100
Expenses:	
Raw materials ($2.40*q*)	42,720
Wages and salaries ($5,200 + $0.30*q*)	10,540
Utilities ($2,400 + $0.05*q*)	3,290
Facility rent ($4,300)	4,300
Insurance ($2,300)	2,300
Miscellaneous ($680 + $0.10*q*)	2,460
Total expense	65,610
Net operating income	$14,490

Required:

1. Calculate the company's activity variances for July. (Hint: Refer to Exhibit 9–6.)
2. Which of the activity variances should be of concern to management? Explain.

EXERCISE 9–3 Revenue and Spending Variances LO9–3

Quilcene Oysteria farms and sells oysters in the Pacific Northwest. The company harvested and sold 8,000 pounds of oysters in August. The company's flexible budget for August appears below:

Quilcene Oysteria
Flexible Budget
For the Month Ended August 31

Actual pounds (*q*)	8,000
Revenue ($4.00*q*)	$ 32,000
Expenses:	
Packing supplies ($0.50*q*)	4,000
Oyster bed maintenance ($3,200)	3,200
Wages and salaries ($2,900 + $0.30*q*)	5,300
Shipping ($0.80*q*)	6,400
Utilities ($830)	830
Other ($450 + $0.05*q*)	850
Total expense	20,580
Net operating income	$ 11,420

The actual results for August appear below:

Quilcene Oysteria Income Statement For the Month Ended August 31	
Actual pounds	8,000
Revenue	$35,200
Expenses:	
Packing supplies	4,200
Oyster bed maintenance	3,100
Wages and salaries	5,640
Shipping	6,950
Utilities	810
Other	980
Total expense	21,680
Net operating income	$13,520

Required:

Calculate the company's revenue and spending variances for August. (Hint: Refer to Exhibit 9–7.)

EXERCISE 9–4 Prepare a Flexible Budget Performance Report LO9–4

Vulcan Flyovers offers scenic overflights of Mount St. Helens, the volcano in Washington State that explosively erupted in 1982. Data concerning the company's operations in July appear below:

Vulcan Flyovers Operating Data For the Month Ended July 31	Actual Results	Flexible Budget	Planning Budget
Flights (q)	48	48	50
Revenue (320.00q$)	$13,650	$15,360	$16,000
Expenses:			
Wages and salaries ($4,000 + 82.00q$)	8,430	7,936	8,100
Fuel (23.00q$)	1,260	1,104	1,150
Airport fees ($650 + 38.00q$)	2,350	2,474	2,550
Aircraft depreciation (7.00q$)	336	336	350
Office expenses ($190 + 2.00q$)	460	286	290
Total expense	12,836	12,136	12,440
Net operating income	$ 814	$ 3,224	$ 3,560

The company measures its activity in terms of flights. Customers can buy individual tickets for overflights or hire an entire plane for an overflight at a discount.

Required:

1. Using Exhibit 9–8 as your guide, prepare a flexible budget performance report for July that includes revenue and spending variances and activity variances.
2. Which of the variances should be of concern to management? Explain.

EXERCISE 9–5 Prepare a Flexible Budget with More Than One Cost Driver LO9–5

Alyeski Tours operates day tours of coastal glaciers in Alaska on its tour boat the Blue Glacier. Management has identified two cost drivers—the number of cruises and the number of passengers—that it uses in its budgeting and performance reports. The company publishes a schedule of day cruises that it may supplement with special sailings if there is sufficient demand. Up to 80 passengers can be accommodated on the tour boat. Data concerning the company's cost formulas appear below:

	Fixed Cost per Month	Cost per Cruise	Cost per Passenger
Vessel operating costs	$5,200	$480.00	$2.00
Advertising	$1,700		
Administrative costs	$4,300	$24.00	$1.00
Insurance	$2,900		

For example, vessel operating costs should be $5,200 per month plus $480 per cruise plus $2 per passenger. The company's sales should average $25 per passenger. In July, the company provided 24 cruises for a total of 1,400 passengers.

Required:
Using Exhibit 9–9 as your guide, prepare the company's flexible budget for July.

EXERCISE 9–6 Critique a Variance Report LO9–6

The Terminator Inc. provides on-site residential pest extermination services. The company has several mobile teams who are dispatched from a central location in company-owned trucks. The company uses the number of jobs to measure activity. At the beginning of April, the company budgeted for 100 jobs, but the actual number of jobs turned out to be 105. A report comparing the budgeted revenues and costs to the actual revenues and costs appears below:

The Terminator Inc.
Variance Report
For the Month Ended April 30

	Actual Results	Planning Budget	Variances
Jobs	105	100	
Revenue	$20,520	$19,500	$ 1,020 F
Expenses:			
Mobile team operating costs	10,320	10,000	320 U
Exterminating supplies	960	1,800	840 F
Advertising	800	800	0
Dispatching costs	2,340	2,200	140 U
Office rent	1,800	1,800	0
Insurance	2,100	2,100	0
Total expense	18,320	18,700	380 F
Net operating income	$ 2,200	$ 800	$ 1,400 F

Required:
Is the above variance report useful for evaluating how well revenues and costs were controlled during April? Why, or why not?

EXERCISE 9–7 Critique a Variance Report LO9–6

Refer to the data for The Terminator Inc. in Exercise 9–6. A management intern has suggested that the budgeted revenues and costs should be adjusted for the actual level of activity in April before they are compared to the actual revenues and costs. Because the actual level of activity was 5% higher than budgeted, the intern suggested that all budgeted revenues and costs should be adjusted upward by 5%. A report comparing the budgeted revenues and costs, with this adjustment, to the actual revenues and costs appears below.

The Terminator Inc.
Variance Report
For the Month Ended April 30

	Actual Results	Adjusted Planning Budget	Variances
Jobs	105	105	
Revenue	$20,520	$20,475	$ 45 F
Expenses:			
Mobile team operating costs	10,320	10,500	180 F
Exterminating supplies	960	1,890	930 F
Advertising	800	840	40 F
Dispatching costs	2,340	2,310	30 U
Office rent	1,800	1,890	90 F
Insurance	2,100	2,205	105 F
Total expense	18,320	19,635	1,315 F
Net operating income	$ 2,200	$ 840	$1,360 F

Required:
Is the above variance report useful for evaluating how well revenues and costs were controlled during April? Why, or why not?

EXERCISE 9–8 Flexible Budgets and Activity Variances LO9–1, LO9–2
Jake's Roof Repair has provided the following data concerning its costs:

	Fixed Cost per Month	Cost per Repair-Hour
Wages and salaries	$23,200	$16.30
Parts and supplies		$8.60
Equipment depreciation	$1,600	$0.40
Truck operating expenses	$6,400	$1.70
Rent	$3,480	
Administrative expenses	$4,500	$0.80

For example, wages and salaries should be $23,200 plus $16.30 per repair hour. The company expected to work 2,800 repair-hours in May, but actually worked 2,900 repair-hours. The company expects its sales to be $44.50 per repair-hour.

Required:
Compute the company's activity variances for May. (Hint: Refer to Exhibit 9–6.)

EXERCISE 9–9 Planning Budget LO9–1
Wyckam Manufacturing Inc. has provided the following information concerning its manufacturing costs:

	Fixed Cost per Month	Cost per Machine-Hour
Direct materials		$4.25
Direct labor	$36,800	
Supplies		$0.30
Utilities	$1,400	$0.05
Depreciation	$16,700	
Insurance	$12,700	

For example, utilities should be $1,400 per month plus $0.05 per machine-hour. The company expects to work 5,000 machine-hours in June. Note that the company's direct labor is a fixed cost.

Required:
Using Exhibit 9–2 as your guide, prepare the company's planning budget for June.

EXERCISE 9–10 Planning Budget LO9–1
Lavage Rapide is a Canadian company that owns and operates a large automatic car wash facility near Montreal. The following table provides data concerning the company's costs:

	Fixed Cost per Month	Cost per Car Washed
Cleaning supplies		$0.80
Electricity	$1,200	$0.15
Maintenance		$0.20
Wages and salaries	$5,000	$0.30
Depreciation	$6,000	
Rent	$8,000	
Administrative expenses	$4,000	$0.10

For example, electricity costs are $1,200 per month plus $0.15 per car washed. The company expects to wash 9,000 cars in August and to collect an average of $4.90 per car washed.

Required:
Using Exhibit 9–2 as your guide, prepare the company's planning budget for August.

EXERCISE 9–11 Flexible Budget LO9–1

Refer to the data for Lavage Rapide in Exercise 9–10. The company actually washed 8,800 cars in August.

Required:

Using Exhibit 9–5 as your guide, prepare the company's flexible budget for August.

EXERCISE 9–12 Activity Variances LO9–2

Refer to the data for Lavage Rapide in Exercise 9–10. The company actually washed 8,800 cars in August.

Required:

Calculate the company's activity variances for August. (Hint: Refer to Exhibit 9–6.)

EXERCISE 9–13 Revenue and Spending Variances LO9–3

Refer to the data for Lavage Rapide in Exercise 9–10. Also assume that the company's actual operating results for August are as follows:

Lavage Rapide Income Statement For the Month Ended August 31	
Actual cars washed	8,800
Revenue	$43,080
Expenses:	
Cleaning supplies	7,560
Electricity	2,670
Maintenance	2,260
Wages and salaries	8,500
Depreciation	6,000
Rent	8,000
Administrative expenses	4,950
Total expense	39,940
Net operating income	$ 3,140

Required:

Calculate the company's revenue and spending variances for August. (Hint: Refer to Exhibit 9–7.)

EXERCISE 9–14 Prepare a Flexible Budget Performance Report LO9–4

Refer to the data for Lavage Rapide in Exercises 9–10 and 9–13.

Required:

Using Exhibit 9–8 as your guide, prepare a flexible budget performance report that shows the company's revenue and spending variances and activity variances for August.

EXERCISE 9–15 Flexible Budget Performance Report in a Cost Center LO9–1, LO9–2, LO9–3, LO9–4

Packaging Solutions Corporation manufactures and sells a wide variety of packaging products. Performance reports are prepared monthly for each department. The planning budget and flexible budget for the Production Department are based on the following formulas, where q is the number of labor-hours worked in a month:

	Cost Formulas
Direct labor	15.80q$
Indirect labor	$8,200 + 1.60q$
Utilities	$6,400 + 0.80q$
Supplies	$1,100 + 0.40q$
Equipment depreciation	$23,000 + 3.70q$
Factory rent	$8,400
Property taxes	$2,100
Factory administration	$11,700 + 1.90q$

The Production Department planned to work 8,000 labor-hours in March; however, it actually worked 8,400 labor-hours during the month. Its actual costs incurred in March are listed below:

	Actual Cost Incurred in March
Direct labor	$134,730
Indirect labor	$19,860
Utilities	$14,570
Supplies	$4,980
Equipment depreciation	$54,080
Factory rent	$8,700
Property taxes	$2,100
Factory administration	$26,470

Required:

1. Using Exhibit 9–2 as your guide, prepare the Production Department's planning budget for the month.
2. Using Exhibit 9–5 as your guide, prepare the Production Department's flexible budget for the month.
3. Using Exhibit 9–8 as your guide, prepare the Production Department's flexible budget performance report for March, including both the spending and activity variances.
4. What aspects of the flexible budget performance report should be brought to management's attention? Explain.

EXERCISE 9–16 Flexible Budgets and Revenue and Spending Variances LO9–1, LO9–3

Via Gelato is a popular neighborhood gelato shop. The company has provided the following cost formulas and actual results for the month of June:

	Fixed Element per Month	Variable Element per Liter	Actual Total for June
Revenue		$12.00	$71,540
Raw materials		$4.65	$29,230
Wages	$5,600	$1.40	$13,860
Utilities	$1,630	$0.20	$3,270
Rent	$2,600		$2,600
Insurance	$1,350		$1,350
Miscellaneous	$650	$0.35	$2,590

While gelato is sold by the cone or cup, the shop measures its activity in terms of the total number of liters of gelato sold. For example, wages should be $5,600 plus $1.40 per liter of gelato sold and the actual wages for June were $13,860. Via Gelato expected to sell 6,000 liters in June, but actually sold 6,200 liters.

Required:

Calculate Via Gelato revenue and spending variances for June. (Hint: Refer to Exhibit 9–7.)

EXERCISE 9–17 Flexible Budget Performance Report LO9–1, LO9–2, LO9–3, LO9–4

AirQual Test Corporation provides on-site air quality testing services. The company has provided the following cost formulas and actual results for the month of February:

	Fixed Component per Month	Variable Component per Job	Actual Total for February
Revenue		$360	$18,950
Technician wages	$6,400		$6,450
Mobile lab operating expenses	$2,900	$35	$4,530
Office expenses	$2,600	$2	$3,050
Advertising expenses	$970		$995
Insurance	$1,680		$1,680
Miscellaneous expenses	$500	$3	$465

The company uses the number of jobs as its measure of activity. For example, mobile lab operating expenses should be $2,900 plus $35 per job, and the actual mobile lab operating expenses for February were $4,530. The company expected to work 50 jobs in February, but actually worked 52 jobs.

Required:

Using Exhibit 9–8 as your guide, prepare a flexible budget performance report showing AirQual Test Corporation's revenue and spending variances and activity variances for February.

EXERCISE 9–18 Working with More Than One Cost Driver LO9–2, LO9–3, LO9–4, LO9–5

The Gourmand Cooking School runs short cooking courses at its small campus. Management has identified two cost drivers it uses in its budgeting and performance reports—the number of courses and the total number of students. For example, the school might run two courses in a month and have a total of 50 students enrolled in those two courses. Data concerning the company's cost formulas appear below:

	Fixed Cost per Month	Cost per Course	Cost per Student
Instructor wages		$3,080	
Classroom supplies			$260
Utilities	$870	$130	
Campus rent	$4,200		
Insurance	$1,890		
Administrative expenses	$3,270	$15	$4

For example, administrative expenses should be $3,270 per month plus $15 per course plus $4 per student. The company's sales should average $800 per student.

The company planned to run three courses with a total of 45 students; however, it actually ran three courses with a total of only 42 students. The actual operating results for September appear below:

	Actual
Revenue	$32,400
Instructor wages	$9,080
Classroom supplies	$8,540
Utilities	$1,530
Campus rent	$4,200
Insurance	$1,890
Administrative expenses	$3,790

Required:

Using Exhibits 9–8 and 9–9 as your guide, prepare a flexible budget performance report that shows both revenue and spending variances and activity variances for September.

Problems

PROBLEM 9–19: Flexible Budget Performance Reports; Working Backwards LO 9–1, LO 9–2, 9–3, 9–4

Ray Company provided the following excerpts from its Production Department's flexible budget performance report:

Ray Company
Production Department Flexible Budget Performance Report
For the Month Ended August 31

	Actual Results	Spending Variances	Flexible Budget	Activity Variances	Planning Budget
Labor-hours (q)	9,480		?		9,000
Direct labor ($?$q$)	$134,730	$?	$132,720	$?	$?
Indirect labor ($? + 1.50q$)	?	1,780 F	21,640	?	?
Utilities ($6,500 + $?$q$)	?	1,450 U	?	336 U	12,800
Supplies ($? + $?$q$)	4,940	?	4,444	?	4,300
Equipment depreciation ($78,400)	?	0	?	?	?
Factory administration ($18,700 + 1.90q$)	?	?	?	?	?
Total expense	$288,088	$?	$?	$?	$?

Required:

Complete the Production Department's Flexible Budget Performance Report by filling in all the question marks.

PROBLEM 9–20 Activity and Spending Variances LO9–1, LO9–2, LO9–3

You have just been hired by FAB Corporation, the manufacturer of a revolutionary new garage door opening device. The president has asked that you review the company's costing system and "do what you can to help us get better control of our manufacturing overhead costs." You find that the company has never used a flexible budget, and you suggest that preparing such a budget would be an excellent first step in overhead planning and control.

After much effort and analysis, you determined the following cost formulas and gathered the following actual cost data for March:

	Cost Formula	Actual Cost in March
Utilities	$20,600 + $0.10 per machine-hour	$24,200
Maintenance	$40,000 + $1.60 per machine-hour	$78,100
Supplies	$0.30 per machine-hour	$8,400
Indirect labor	$130,000 + $0.70 per machine-hour	$149,600
Depreciation	$70,000	$71,500

During March, the company worked 26,000 machine-hours and produced 15,000 units. The company had originally planned to work 30,000 machine-hours during March.

Required:

1. Calculate the activity variances for March. (Hint: Refer to Exhibit 9–6.) Explain what these variances mean.
2. Calculate the spending variances for March. (Hint: Refer to Exhibit 9–7.) Explain what these variances mean.

PROBLEM 9–21 More than One Cost Driver LO9–2, LO9–3, LO9–4, LO9–5

Milano Pizza is a small neighborhood pizzeria that has a small area for in-store dining as well as offering take-out and free home delivery services. The pizzeria's owner has determined that the shop has two major cost drivers—the number of pizzas sold and the number of deliveries made.

The pizzeria's cost formulas appear below:

	Fixed Cost per Month	Cost per Pizza	Cost per Delivery
Pizza ingredients		$3.80	
Kitchen staff	$5,220		
Utilities	$630	$0.05	
Delivery person			$3.50
Delivery vehicle	$540		$1.50
Equipment depreciation	$275		
Rent	$1,830		
Miscellaneous	$820	$0.15	

In November, the pizzeria budgeted for 1,200 pizzas at an average selling price of $13.50 per pizza and for 180 deliveries.

Data concerning the pizzeria's actual results in November appear below:

	Actual Results
Pizzas	1,240
Deliveries	174
Revenue	$17,420
Pizza ingredients	$4,985
Kitchen staff	$5,281
Utilities	$984
Delivery person	$609
Delivery vehicle	$655
Equipment depreciation	$275
Rent	$1,830
Miscellaneous	$954

Required:

1. Using Exhibits 9–8 and 9–9 as your guide, prepare a flexible budget performance report that shows both revenue and spending variances and activity variances for the pizzeria for November.
2. Explain the activity variances.

PROBLEM 9–22 Critique a Report; Prepare a Performance Report LO9–1, LO9–2, LO9–3, LO9–4, LO9–6

TipTop Flight School offers flying lessons at a small municipal airport. The school's owner and manager has been attempting to evaluate performance and control costs using a variance report that compares the planning budget to actual results. A recent variance report appears below:

TipTop Flight School
Variance Report
For the Month Ended July 31

	Actual Results	Planning Budget	Variances	
Lessons	155	150		
Revenue	$33,900	$33,000	$900	F
Expenses:				
Instructor wages	9,870	9,750	120	U
Aircraft depreciation	5,890	5,700	190	U
Fuel	2,750	2,250	500	U
Maintenance	2,450	2,330	120	U
Ground facility expenses	1,540	1,550	10	F
Administration	3,320	3,390	70	F
Total expense	25,820	24,970	850	U
Net operating income	$ 8,080	$ 8,030	$ 50	F

After several months of using such variance reports, the owner has become frustrated. For example, she is quite confident that instructor wages were very tightly controlled in July, but the report shows an unfavorable variance.

The planning budget was developed using the following formulas, where q is the number of lessons sold:

	Cost Formulas
Revenue	220q$
Instructor wages	65q$
Aircraft depreciation	38q$
Fuel	15q$
Maintenance	$530 + 12q$
Ground facility expenses	$1,250 + 2q$
Administration	$3,240 + 1q$

Required:

1. Should the owner feel frustrated with the variance reports? Explain.
2. Using Exhibit 9–8 as your guide, prepare a flexible budget performance report for the school for July.
3. Evaluate the school's performance for July.

PROBLEM 9–23 Performance Report for a Nonprofit Organization LO9–1, LO9–2, LO9–3, LO9–4, LO9–6

The St. Lucia Blood Bank, a private charity partly supported by government grants, is located on the Caribbean island of St. Lucia. The blood bank has just finished its operations for September, which was a very busy month due to a powerful hurricane that hit neighboring islands causing many injuries. The hurricane largely bypassed St. Lucia, but residents of St. Lucia willingly donated their blood to help people on other islands. As a consequence, the blood bank collected and processed over 20% more blood than had been originally planned for the month.

A report prepared by a government official comparing actual costs to budgeted costs for the blood bank appears below. Continued support from the government depends on the blood bank's ability to demonstrate control over its costs.

St. Lucia Blood Bank
Cost Control Report
For the Month Ended September 30

	Actual Results	Planning Budget	Variances
Liters of blood collected	620	500	
Medical supplies	$ 9,250	$ 7,500	$1,750 U
Lab tests	6,180	6,000	180 U
Equipment depreciation	2,800	2,500	300 U
Rent	1,000	1,000	0
Utilities	570	500	70 U
Administration	11,740	11,250	490 U
Total expense	$31,540	$28,750	$2,790 U

The managing director of the blood bank was very unhappy with this report, claiming that his costs were higher than expected due to the emergency on the neighboring islands. He also pointed out that the additional costs had been fully covered by payments from grateful recipients on the other islands. The government official who prepared the report countered that all of the figures had been submitted by the blood bank to the government; he was just pointing out that actual costs were a lot higher than promised in the budget.

The following cost formulas were used to construct the planning budget:

	Cost Formulas
Medical supplies	$\$15.00q$
Lab tests	$\$12.00q$
Equipment depreciation	$2,500
Rent	$1,000
Utilities	$500
Administration	$\$10{,}000 + \$2.50q$

Required:

1. Using Exhibit 9–8 as your guide, prepare a flexible budget performance report for September.
2. Do you think any of the variances in the report you prepared should be investigated? Why?

PROBLEM 9–24 Critiquing a Report; Preparing a Performance Budget LO9–1, LO9–2, LO9–3, LO9–4, LO9–6

Exchange Corp. is a company that acts as a facilitator in tax-favored real estate swaps. Such swaps, know as 1031 exchanges, permit participants to avoid some or all of the capital gains taxes that would otherwise be due. The bookkeeper for the company has been asked to prepare a report for the company to help its owner/manager analyze performance. The first such report appears below:

Exchange Corp.
Analysis of Revenues and Costs
For the Month Ended May 31

	Actual Unit Revenues and Costs	Planning Budget Unit Revenues and Costs	Variances
Exchanges completed	50	40	
Revenue	$385	$395	$10 U
Expenses:			
Legal and search fees	184	165	19 U
Office expenses	112	135	23 F
Equipment depreciation	8	10	2 F
Rent	36	45	9 F
Insurance	4	5	1 F
Total expense	344	360	16 F
Net operating income	$ 41	$ 35	$ 6 F

Note that the revenues and costs in the above report are *unit* revenues and costs. For example, the average office expense is $135 per exchange completed on the planning budget; whereas, the average actual office expense is $112 per exchange completed.

Legal and search fees is a variable cost; office expenses is a mixed cost; and equipment depreciation, rent, and insurance are fixed costs. In the planning budget, the fixed component of office expenses was $5,200.

All of the company's revenues come from fees collected when an exchange is completed.

Required:

1. Evaluate the report prepared by the bookkeeper.
2. Using Exhibit 9–8 as your guide, prepare a performance report that would help the owner/manager assess the performance of the company in May.
3. Using the report you created, evaluate the performance of the company in May.

PROBLEM 9–25 Critiquing a Variance Report; Preparing a Performance Report LO9–1, LO9–2, LO9–3, LO9–4, LO9–6

Several years ago, Westmont Corporation developed a comprehensive budgeting system for planning and control purposes. While departmental supervisors have been happy with the system, the factory manager has expressed considerable dissatisfaction with the information being generated by the system.

A report for the company's Assembly Department for the month of March follows:

Assembly Department
Cost Report
For the Month Ended March 31

	Actual Results	Planning Budget	Variances
Machine-hours	35,000	40,000	
Variable costs:			
Supplies	$ 29,700	$ 32,000	$2,300 F
Scrap	19,500	20,000	500 F
Indirect materials	51,800	56,000	4,200 F
Fixed costs:			
Wages and salaries	79,200	80,000	800 F
Equipment depreciation	60,000	60,000	—
Total cost	$240,200	$248,000	$7,800 F

After receiving a copy of this cost report, the supervisor of the Assembly Department stated, "These reports are super. It makes me feel really good to see how well things are going in my department. I can't understand why those people upstairs complain so much about the reports."

For the last several years, the company's marketing department has chronically failed to meet the sales goals expressed in the company's monthly budgets.

Required:

1. The company's president is uneasy about the cost reports and would like you to evaluate their usefulness to the company.
2. What changes, if any, should be made in the reports to give better insight into how well departmental supervisors are controlling costs?
3. Using Exhibit 9–8 as your guide, prepare a new performance report for the quarter, incorporating any changes you suggested in question (2) above.
4. How well were costs controlled in the Assembly Department in March?

PROBLEM 9–26 Critiquing a Cost Report; Preparing a Performance Report LO9–1, LO9–2, LO9–3, LO9–4, LO9–6

Frank Weston, supervisor of the Freemont Corporation's Machining Department, was visibly upset after being reprimanded for his department's poor performance over the prior month. The department's cost control report is given below:

Freemont Corporation–Machining Department Cost Control Report For the Month Ended June 30	Actual Results	Planning Budget	Variances
Machine-hours	38,000	35,000	
Direct labor wages	$ 86,100	$ 80,500	$ 5,600 U
Supplies	23,100	21,000	2,100 U
Maintenance	137,300	134,000	3,300 U
Utilities	15,700	15,200	500 U
Supervision	38,000	38,000	0
Depreciation	80,000	80,000	0
Total	$380,200	$368,700	$11,500 U

"I just can't understand all of these unfavorable variances," Weston complained to the supervisor of another department. "When the boss called me in, I thought he was going to give me a pat on the back because I know for a fact that my department worked more efficiently last month than it has ever worked before. Instead, he tore me apart. I thought for a minute that it might be over the supplies that were stolen out of our warehouse last month. But they only amounted to a couple of hundred dollars, and just look at this report. Everything is unfavorable."

Direct labor wages and supplies are variable costs; supervision and depreciation are fixed costs; and maintenance and utilities are mixed costs. The fixed component of the budgeted maintenance cost is $92,000; the fixed component of the budgeted utilities cost is $11,700.

Required:

1. Evaluate the company's cost control report and explain why the variances were all unfavorable.
2. Using Exhibit 9–8 as your guide, prepare a performance report that will help Mr. Weston's superiors assess how well costs were controlled in the Machining Department.

connect Cases

CASE 9–27 Ethics and the Manager LO9–3

Tom Kemper is the controller of the Wichita manufacturing facility of Prudhom Enterprises, Inc.

The annual cost control report is one of the many reports that must be filed with corporate headquarters and is due at corporate headquarters shortly after the beginning of the New Year. Kemper does not like putting work off to the last minute, so just before Christmas he prepared a preliminary draft of the cost control report. Some adjustments would later be required for transactions that occur between Christmas and New Year's Day. A copy of the preliminary draft report, which Kemper completed on December 21, follows:

Wichita Manufacturing Facility Cost Control Report December 21 Preliminary Draft	Actual Results	Flexible Budget	Spending Variances
Labor-hours	18,000	18,000	
Direct labor	$ 326,000	$ 324,000	$ 2,000 U
Power	19,750	18,000	1,750 U
Supplies	105,000	99,000	6,000 U
Equipment depreciation	343,000	332,000	11,000 U
Supervisory salaries	273,000	275,000	2,000 F
Insurance	37,000	37,000	0
Industrial engineering	189,000	210,000	21,000 F
Factory building lease	60,000	60,000	0
Total expense	$1,352,750	$1,355,000	$ 2,250 F

Melissa Ilianovitch, the general manager at the Wichita facility, asked to see a copy of the preliminary draft report. Kemper carried a copy of the report to her office where the following discussion took place:

Ilianovitch: Wow! Almost all of the variances on the report are unfavorable. The only favorable variances are for supervisory salaries and industrial engineering. How did we have an unfavorable variance for depreciation?

Kemper: Do you remember that milling machine that broke down because the wrong lubricant was used by the machine operator?

Ilianovitch: Yes.

Kemper: We couldn't fix it. We had to scrap the machine and buy a new one.

Ilianovitch: This report doesn't look good. I was raked over the coals last year when we had just a few unfavorable variances.

Kemper: I'm afraid the final report is going to look even worse.

Ilianovitch: Oh?

Kemper: The line item for industrial engineering on the report is for work we hired Ferguson Engineering to do for us. The original contract was for $210,000, but we asked them to do some additional work that was not in the contract. We have to reimburse Ferguson Engineering for the costs of that additional work. The $189,000 in actual costs that appears on the preliminary draft report reflects only their billings up through December 21. The last bill they had sent us was on November 28, and they completed the project just last week. Yesterday I got a call from Laura Sunder over at Ferguson and she said they would be sending us a final bill for the project before the end of the year. The total bill, including the reimbursements for the additional work, is going to be . . .

Ilianovitch: I am not sure I want to hear this.

Kemper: $225,000

Ilianovitch: Ouch!

Kemper: The additional work added $15,000 to the cost of the project.

Ilianovitch: I can't turn in a report with an overall unfavorable variance! They'll kill me at corporate headquarters. Call up Laura at Ferguson and ask her not to send the bill until after the first of the year. We have to have that $21,000 favorable variance for industrial engineering on the report.

Required:

What should Tom Kemper do? Explain.

CASE 9–28 Critiquing a Report; Calculating Spending Variances LO9–3, LO9–5, LO9–6

Boyne University offers an extensive continuing education program in many cities throughout the state. For the convenience of its faculty and administrative staff and to save costs, the university operates a motor pool. The motor pool's monthly planning budget is based on operating 20 vehicles; however, for the month of March the university purchased one additional vehicle. The motor pool furnishes gasoline, oil, and other supplies for its automobiles. A mechanic does routine maintenance and minor repairs. Major repairs are performed at a nearby commercial garage.

The following cost control report shows actual operating costs for March of the current year compared to the planning budget for March.

Boyne University Motor Pool
Cost Control Report
For the Month Ended March 31

	March Actual	Planning Budget	(Over) Under Budget
Miles	63,000	50,000	
Autos	21	20	
Gasoline	$ 9,350	$ 7,500	$(1,850)
Oil, minor repairs, parts	2,360	2,000	(360)
Outside repairs	1,420	1,500	80
Insurance	2,120	2,000	(120)
Salaries and benefits	7,540	7,540	0
Vehicle depreciation	5,250	5,000	(250)
Total	$28,040	$25,540	$(2,500)

The planning budget was based on the following assumptions:

a. $0.15 per mile for gasoline.
b. $0.04 per mile for oil, minor repairs, and parts.
c. $75 per automobile per month for outside repairs.
d. $100 per automobile per month for insurance.
e. $7,540 per month for salaries and benefits.
f. $250 per automobile per month for depreciation.

The supervisor of the motor pool is unhappy with the report, claiming it paints an unfair picture of the motor pool's performance.

Required:

1. Calculate the spending variances for March. (Hint: Refer to Exhibit 9–7.)
2. What are the deficiencies in the original cost control report? How do your calculations in part (1) above overcome these deficiencies?

(CMA, adapted)

CASE 9–29 Performance Report with More than One Cost Driver LO9–1, LO9–2, LO9–3, LO9–4, LO9–5

The Little Theatre is a nonprofit organization devoted to staging plays for children. The theater has a very small full-time professional administrative staff. Through a special arrangement with the actors' union, actors and directors rehearse without pay and are paid only for actual performances.

The Little Theatre had tentatively planned to put on six different productions with a total of 108 performances. For example, one of the productions was *Peter Rabbit,* which had a six-week run with three performances on each weekend. The costs from the current year's planning budget appear below.

The Little Theatre Costs from the Planning Budget For the Year Ended December 31	
Budgeted number of productions	6
Budgeted number of performances	108
Actors and directors wages	$216,000
Stagehands wages	32,400
Ticket booth personnel and ushers wages	16,200
Scenery, costumes, and props	108,000
Theater hall rent	54,000
Printed programs	27,000
Publicity	12,000
Administrative expenses	43,200
Total	$508,800

Some of the costs vary with the number of productions, some with the number of performances, and some are fixed and depend on neither the number of productions nor the number of performances. The costs of scenery, costumes, props, and publicity vary with the number of productions. It doesn't make any difference how many times *Peter Rabbit* is performed, the cost of the scenery is the same. Likewise, the cost of publicizing a play with posters and radio commercials is the same whether there are 10, 20, or 30 performances of the play. On the other hand, the wages of the actors, directors, stagehands, ticket booth personnel, and ushers vary with the number of performances. The greater the number of performances, the higher the wage costs will be. Similarly, the costs of renting the hall and printing the programs will vary with the number of performances. Administrative expenses are more difficult to analyze, but the best estimate is that approximately 75% of the budgeted costs are fixed, 15% depend on the number of productions staged, and the remaining 10% depend on the number of performances.

After the beginning of the year, the board of directors of the theater authorized expanding the theater's program to seven productions and a total of 168 performances. Not surprisingly, actual costs were considerably higher than the costs from the planning budget. (Grants from donors and ticket sales were also correspondingly higher, but are not shown here.) Data concerning the actual costs appear below:

The Little Theatre Actual Costs For the Year Ended December 31	
Actual number of productions	7
Actual number of performances	168
Actors and directors wages	$341,800
Stagehands wages	49,700
Ticket booth personnel and ushers wages	25,900
Scenery, costumes, and props	130,600
Theater hall rent	78,000
Printed programs	38,300
Publicity	15,100
Administrative expenses	47,500
Total	$726,900

Required:

1. Use Exhibit 9–8 as your guide, prepare a flexible budget performance report for the year that shows both spending variances and activity variances.
2. If you were on the board of directors of the theater, would you be pleased with how well costs were controlled during the year? Why, or why not?
3. The cost formulas provide figures for the average cost per production and average cost per performance. How accurate do you think these figures would be for predicting the cost of a new production or of an additional performance of a particular production?